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Integrating operations and marketing decisions to manage perishability risks with target minimum remaining shelf-life available to consumers

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ABSTRACT

We study an aggregate level planning problem at the bottleneck packaging stage of a Make-and-Pack production system of perishable products and integrate sequencing decisions. The perishability risks are accounted for by producing to order with remaining shelf-life (RSL) dependent demand and age-dependent expected product holding cost till dispatch. Products dispatch time determines the Maximum-RSL delivered. Unlike past studies, we examine a target Minimum-RSL that marketing aims to provide. The product demand decreases from its Maximum-RSL to the Minimum-RSL. The problem is modeled as a two-stage stochastic 0–1 integer non-linear programming problem. The first-stage decisions are the inventory of intermediate products, packaging sequence, capacity allocation, Maximum-RSL, and Minimum-RSL. In the second stage, market demand is realized and has a stochastic error component. The stochastic problem is converted into its deterministic equivalent using the Sample-Average-Approximation method. Representing the demand scenarios using Low Discrepancy Sequences and reducing variance using the Antithetic Variates technique provide stable solutions. Sensitivity analysis demonstrates that the integrated-firm improves freshness criteria (Maximum-RSL and Minimum-RSL) by reducing setup time/demand variability/demand satisfaction requirement (shortage cost) or increasing importance of customer service in product freshness (product holding cost). Maximum-RSL increases with increased packaging rate. High demand variability, shortage cost, product holding cost, and low packaging rate promote integrated sequencing-capacity allocation. An efficient heuristic based on the Generalized Benders' Decomposition method is proposed to solve large-size problems. The heuristic provides a better solution than the DICOPT solver, solves far fewer non-linear programming problems than the Simple Branch-and-Bound solver, with less computational effort.

1. Introduction

A product is termed perishable if its physical condition worsens with time, and its value reduces in the customer's perception (Raafat, 1991). The shelf-life of a food product is the time-span until it is safe to use, and its physical, chemical, and microbiological characteristics retain and comply with the nutritional standard. An average of 35 percent of food products is wasted from production to consumption (Kaipia et al., 2013).

Customers check the expiration date because of their awareness about the risk associated with the perishable product. Customers believe that a high remaining shelf-life indicates good quality and determines their product choice (Tsiros & Heilman, 2005). Usually, customers purchase food items to consume them, say, in the coming week. They require a remaining shelf-life then of at least a week at the time of purchase. The remaining shelf-life is the remaining time until the end of the product shelf-life. Therefore, the retailer has a target minimum remaining shelf-life (Min-RSL) they wish to provide to the consumers (Van Donselaar et al., 2006).

In several situations of the perishable goods supply chain, decisions on production scheduling at manufacture and inventory holding at the warehouse are coordinated to meet demand at the distributor/retailer (Masruroh et al., 2020). Also, a firm may supply perishable products directly to the store (point-of-sale) and bypass the retailer's warehouses (saving time and physical handling) (Amorim et al., 2014). A combination of several factors, including higher availability of raw-material, better utilization of processing unit, improved distribution, and less wastage, leads to products with higher remaining shelf-life at the retailer with a positive impact on consumption (Castrica et al., 2021).

Retailers order close to the amount they expect to sell, and in the case of perishables, it is not unusual for them to order less than the sales

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forecast. They then experience stockout or close to stockout by the time the next replenishment arrives. Overstocking risk is managed by the retailer in several ways, for example, by selling at a discount (Buisman et al., 2019, Azadi et al., 2019) or donating to the food bank (Krommyda et al., 2020) or returning to the supplier (Chen et al. 2020), the products whose remaining shelf-life is less than the Min-RSL.

1.1. Research motivation

Several perishable products, e.g., yogurt, are produced in a Make-and-Pack production system (Lütke Entrup et al., 2005; Günther et al., 2006). The Make-and-Pack production system consists of a single production stage and, subsequently, the packaging stage. Challenges like production constraints (e.g., setup requirement), product characteristics (e.g., perishability), and customer's behavior need to be addressed during production planning of a perishable product. The quality of the product deteriorates continuously over time, and this aspect has to be incorporated in production planning. Any delay in selling the product, after production, results in shelf-life loss and marketability loss.

This work was motivated by the industrial situation of yogurt processing described in Yildiz (2016), Bylund (1995), Chandan et al. (2009), Entrup (2006) and Kopanos et al. (2010) and which is summarized in Table 1. In the case of yogurt, the Make-stage consists of a preparation stage and fermentation stage. The raw milk preparation consists of standardization, pasteurization, and homogenization processes in sequence. After preparation, the raw milk is fermented, and the intermediate product is stored in a buffer tank. In the packaging stage, flavoring (if required), filling, and packaging take place. In the flavoring stage, sugar and fruits are added to make further yogurt types. The products are filled, packed, and dispatched to retailers.

Studies tend to prioritize production planning at the packaging stage since it is a bottleneck resource and capital-intensive (Gellert et al., 2011). The bottleneck at the packaging stage occurs due to a low packaging rate compared to the previous stage's flow rate and the product diversity in terms of flavors and packaging sizes (Amorim et al., 2011). At the packaging line, a change in an intermediate product requires a major setup (cleaning and sterilizing the packaging line), and a change in packet sizes, materials, and labels requires a minor setup.

In developed countries, e.g., the United Kingdom, retailers, due to their higher concentration, have more negotiating power than the yogurt processors (McLaughlin, 2002). The profit-margin for yogurt is marginal, and it is increasingly vital for yogurt processors to differentiate themselves. One way yogurt marketers can differentiate themselves is in terms of (1) the maximum remaining shelf-life (Max-RSL) delivered to the retailer (and available to consumers) and (2) the Min-RSL (which is strictly less than the Max-RSL).

Table1
Production environment.

	Fermentation stage
Product	The fermented coagulum of yogurt.
Major setup	The changeover among different fat levels requires cleaning the process tank, and it takes at most 2 h.
Capacity	The fermentation capacity is in liter per day, and it depends on the size of the process tank and the time required for fermentation.
Process	Standardized and pasteurized milk filled in process tanks and starting culture is added after some minutes of agitation.
Time	The time required to ferment is generally 4–6 h, but it varies significantly depending on culture concentration, temperature, and the kind of product texture required.
	Flavoring and Packaging stage
Products	Fruit, sweet and sour yogurt in different packet sizes.
Major setup	Changeover among different flavors and different fat levels requires cleaning of the packaging line.
Minor setup	Format change and retuning of the packaging line are required when the changeover occurs among different packet sizes of the same flavor and same fat level.
Capacity	Packaging capacity is in packets/units per hour.
Process	Yogurt filled and packed.

1.2. Research objective and approach

The objective of this study is to determine aggregate level operations decisions on (i) starting inventory of each intermediate product, (ii) length of the production planning cycle, which is the period time between consecutive dispatches, (iii) capacity allocation to each product, and the marketing/retailers decision on the Min-RSL. The packaging sequence decision of products is integrated with these decisions.

The marketing plan for delivering products and services shall fail without proper coordination with the operations (Tang, 2010). Operations determine the length of the production planning cycle and thereby the Max-RSL provided to the retailer. Marketing determines the Min-RSL. A common dispatch time and the operations decision to pack in larger lots increases dispatch time and shelf-life loss before dispatch. The packaging sequence of products also impacts the shelf-life loss. A lower Min-RSL implies the pursuit of higher sales but at the risk of overstocking at the retailer. A shorter production planning cycle (i.e., the period time between consecutive deliveries) shall lead to higher Max-RSL. It helps to achieve the marketing's objective of customer service in providing fresh products but implies a frequent changeover on the packaging line, smaller lots, and higher operational cost.

The problem is modeled as a two-stage stochastic 0–1 integer non-linear programming (SMINLP) problem for a pre-defined packaging sequence. In the first stage, operations and marketing decisions are taken. The first stage decisions are taken before demand realization, and the second stage decisions are taken after. The second stage's product demand decreases from the time of its dispatch/delivery (the Max-RSL) to the Min-RSL. The period time over which demand is estimated is limited by the length of the production planning cycle. The demand function has a stochastic error component to capture uncertainty. In the second stage, the intermediate product inventory is allocated to execute the packaging plan. The stochastic problem is converted into its deterministic equivalent using Sample Average Approximation (SAA) method. Representing the demand scenarios using Low Discrepancy Sequences (LDS) (Sobol, 1976) and reducing variance using the Antithetic Variates technique (Law et al., 2000) provide stable solutions (Kuthambalayan et al., 2015). All possible sequences are evaluated to identify the optimal sequence.

The proposed mathematical programming model allows operations managers to gain insights into an integrated firm's ability to balance the conflicting objectives of reducing costs and improving customer service in terms of product freshness. The model can be used to identify the conditions promoting integrated sequencing and capacity allocation and examine the key trade-offs between Max-RSL and Min-RSL. Numerical comparisons provide insights into the benefits of joint decision-making and conditions in which optimal packaging sequence choice provides significant benefit. We compute the optimal stocking level of intermediate products, length of the production planning cycle, packaging sequence, capacity allocation, and Min-RSL in numerical experiments. The qualitative insights are based on extensive numerical experiments. The pattern of results obtained is reported as observations (Swaminathan & Tayur, 1998; Gupta & Benjaafar, 2004; Altendorfer & Minner, 2014; Kuthambalayan et al., 2014; Kuthambalayan & Bera, 2020).

We propose an efficient heuristic based on the Generalized Benders' Decomposition procedure (Benders', 1962; Geoffrion, 1972) for solving a reformulated model which permits the change in production sequence with the change in demand scenarios. The MINLP problem is decomposed into a non-standard Master and Sub-problem. The proposed heuristic is evaluated on several problem instances, along with commercial solvers, the Simple Branch-and-Bound (SBB) and, the DICOPT. The SBB solver implements the Non-linear Branch-and-Bound method (Gupta & Ravindran, 1985), and the DICOPT solver implements Outer-Approximation method (Duran & Grossmann, 1986). Like the GBD

method, DICOPT solves a mixed-integer programming (MIP) problem and a non-linear programming (NLP) problem at each iteration. Several large problem instances are solved, and the Worst, Average, and Best gap between the Benders' Heuristic method and SBB solutions and DICOPT and SBB solutions are computed. The number of NLP solves and computational effort observed.

The remainder of the paper is organized as follows. Section 2 provides the related literature and positions our work in the literature. Section 3 describes the notations and assumptions used throughout this study and present both the mathematical models. Section 4 describes the solution approach to each of the models. In Section 5, a numerical example is provided. The effects of key parameters are studied to illustrate the benefit of integrated decisions at the Operations-Marketing interface and simultaneous sequencing and aggregate level planning. Section 5 also demonstrates the efficacy of the proposed solution technique to solve the reformulated model of practical size. Finally, in Section 6, we draw some conclusions.

2. Literature review

Wei et al. (2019) incorporate perishability issues, considering shelf lives of intermediates as well as raw materials and end products, by extending multi-level lot-sizing and scheduling problem formulation. They demonstrate that classical models' production plans are often infeasible under a setting with perishable products.

2.1. Studies focusing on production planning of perishable goods

Studies focusing on the packaging stage of the Make-and-Pack production of perishable products aim to analyze the trade-offs between cost and customer service in the design of production systems and address challenges in improving these systems. They modify the traditional mathematical models to account for perishability risks in the formulation explicitly.

The shelf-life limitations of perishable products are integrated with production planning and scheduling in a diverse manner by using**: (1)** Make-to-Order (MTO) production strategy, **(2)(a)** Higher inventory costs for products with shorter shelf-life or **(b)** age-dependent inventory holding cost, **(3)** Penalty in objective function on units of product which remain unsold beyond their shelf-life, **(4)(a)** FIFO inventory issue or **(b)** mixed FIFO and LIFO inventory issue (with micro-period age-dependent inventory policy), **(5)** A constraint on stocking period, **(6)** Remaining shelf-life dependent price, **(7)** Remaining shelf-life dependent demand, **(8)** Multi-objective problem formulation with **(a)** maximizing the remaining shelf-life at delivery as one of the objectives or **(b)** minimization of variability in expected profit as one of the objectives, **(9)** Product deterioration rate, and **(10)** Constraint on product storage capacity. The related works (which is not exhaustive) are provided in Table 2.

Lütke Entrup et al. (2005) address the problem of generating financial benefit by providing additional remaining shelf-life to customers and balancing the associated costs (including overtime cost). The customers have a minimum remaining shelf-life requirement.

Marinelli et al. (2007) employ a MTO approach without shortages (also, there are no backorders and starting, ending inventories). They consider a shared buffer of limited capacity. The problem is a capacitated lot sizing and scheduling problem (CLSP) with a capacity increment cost. They propose a heuristic procedure to solve the problem.

Kopanos et al. (2010) address the problem of product families' assignment to processing units and determine their production sequencing considering overtime cost. The model also determines the production quantity and inventory level of each product in a family.

Amorim et al. (2011) consider the mixed Make-to-Stock (MTS)-MTO strategy, besides the MTO production strategy, to reduce changeovers and reduce costs. They use a multi-objective problem formulation. The problem is solved using the Genetic Algorithm and a MIP solver.

Amorim et al. (2013) and Amorim et al. (2014) use a two-stage stochastic production planning model to manage perishable food products. The first stage decisions are production lots and the setup schedule of the pre-defined production sequence. Amorim et al. (2013) evaluate model extensions that produce plans with low variability and high expected profit. Amorim et al. (2014) consider direct supply to store. Santos et al. (2019) determine productions decision before the planning horizon starts, and the inventory levels (total production minus the demand of the scenario), backlog and spoiled production after the demand realization considering FIFO inventory policy. They reformulate the robust problem as a non-linear problem. Using classical techniques, the problem is linearized and solved using the row-and-column generation algorithm.

Sel et al. (2015) and Escobet et al. (2019) consider shelf-life limitations in a constraint programming model to batch schedule and integrate planning-scheduling of yogurt production. Each day demand has a due-date, and unmet demand is lost. Sel et al. (2015) propose a decomposition heuristic to solve the MIP problem. Escobet et al. (2019) use multi-objective formulation (which minimizes production span, cost of resources, and relaxation in the task starting and ending time).

Soler et al. (2019) address a lot-sizing and scheduling problem in which the plant layout change and creation of temporary workstations may be required to assemble the production lines. Each of the perishable products can be produced only by a particular production line. They solve the MIP problem using the branch-and-bound algorithm and propose improved branching rules and a relax-and-fit procedure.

Table 2
Integrating shelf-life limitations of perishable products with production planning.

	1	2a	2b	3	4a	4b	5	6	7	8a	8b	9	10
Lütke Entrup et al. (2005)							✓	✓					
Marinelli et al. (2007)	✓												
Kopanos et al. (2010)		✓											
Amorim et al. (2011)							✓			✓			
Amorim et al. (2013)				✓					✓		✓		
Amorim et al. (2014)				✓					✓				
Sel et al. (2015)	✓												
Janssen et al. (2018)						✓							
Zandieh & Motallebi (2018)	✓												
Escobet et al. (2019)	✓												
Soler et al. (2019)								✓					
Nobil et al. (2019)				✓								✓	
Georgiadis et al. (2020)		✓											
Alipour et al. (2020)				✓				✓					
Acevedo-Ojeda et al. (2020)				✓									
Santos et al. (2019)				✓	✓								
Jing & Chao (2021)			✓									✓	✓
Our work	✓		✓						✓				

Alipour et al. (2020) propose a relax-and-fix and fix-and-optimize procedure to solve the model for lot-sizing and scheduling. Georgiadis et al. (2020) study the lot-sizing and production scheduling problem with the rescheduling aspect in a rolling horizon. Any new information such as new order, order modification and/or cancellation, and deviation of actual to planned production is utilized in rescheduling.

The order penetration point is often at the intermediate product. Zandieh & Motallebi (2018), using discrete-event simulation, recommend planning intermediate product inventory based on forecast and packaging based on firm orders. Certain studies focus to assess the impact of intermediate product perishability on production of higher-level items. Sel et al. (2017) study the yogurt planning and scheduling problem with uncertainty in quality decay of intermediate mixture using a simulation approach. Acevedo-Ojeda et al. (2020) study four variants of the two-level lot-sizing problem. They incorporate different types of intermediate product perishability: (a) fixed shelf-life, (b) functionality deterioration with time, (c) volume deterioration (i.e., loss of material) with time, and (d) functionality-volume deterioration.

Nobil et al. (2019) determine the optimal values for period length and production lot size by minimizing the manufacturing cost, carrying cost, machine setup cost, and disposal costs of the items (which deteriorate at a negative exponential rate) using the bi-section (root-finding) method and compare the calculation effort with the Newton-Rapson method.

Perishable products need suitable temperature and humidity for storage. Jing & Chao (2021) extend the conventional Wagner-Whitin's Dynamic Lot-sizing model and its variations by assuming age-dependent inventory holding costs in every given period, deterioration rate, and stockout (due to high setup costs and limitation on storage capacity), and use dynamic programming to solve.

Janssen et al. (2018) study micro-period (based on time of day -morning, midday, afternoon, or evening) inventory policy under random demand, target customer service-level, and lost sales. A mixed FIFO and LIFO issuing policy is considered with stock classified based on the age of items.

2.2. Research positioning

In this study, the limited shelf-life consideration of products is integrated into aggregate level production planning, along with sequencing, by using the MTO production strategy in which the aggregate level demand for a product is considered to decreases with a decrease in remaining shelf-life. A product's shelf-life loss is computed from the start of setup to encourage a reduction in setup time and production in small lots. A holding cost, which is equivalent for each unit in the lot, is imposed from its setup until its dispatch. This period is the shelf-life loss at dispatch. Small product lots packed close to dispatch time incurs lower holding costs.

In several of the studies reviewed,

1. The planning horizon is of finite length (e.g., Week, Month) and divided into periods. In our case, the production cycle is a decision variable.
2. The product demand is deterministic but dynamic. In our case, the product demand is stochastic and stationary.
3. Setups are sequence-independent between families and sequence-dependent for products within the family. In our case, for aggregate planning at the product family level, only sequence-independent setups are considered.
4. There is no age-dependent expected product holding cost component in the objective function, unlike ours.
5. Minimum lot-sizes (causing production in excess of demand) and/or maximum lot-sizes (based on the previous stage's supply capacity) are enforced and product demand estimated till the end of its shelf-life. In our case, the supply is only limited by the product demand which is estimated more conservatively based on the Max-RSL at the time of dispatch till the Min-RSL. This period is restricted by the time between consecutive dispatch/production cycle length, which is limited by the shelf-life. The few studies which consider Min-RSL, e.g., Lütke Entrup et al. (2005), fix its value.

The modeling approach jointly decides the stocking level of the intermediate product, packaging sequence, capacity allocation, time of dispatch, and the Min-RSL. The operation's objective of cost reduction is thus balanced with the marketing objective of providing fresh products by integrated-decisions on batch sizes, length of production planning cycles (frequency of dispatch), and the Min-RSL. The purpose is to evaluate the integrated operations-marketing firm's ability to balance two conflicting objectives: reducing costs and improving customer service in terms of product freshness. Sequencing and capacity allocation can be performed in one single or two separate planning steps. However, since both decisions impact product demand, integrated planning shall prove beneficial. We also aim to gain insights into conditions that promote the integration of sequencing and capacity allocation decisions.

In several of the studies reviewed, the problem for a pre-defined production sequence is modeled as a MIP problem. We propose a SMINLP model and use the SAA method to convert it into its deterministic equivalent. The demand scenarios are generated using Sobols's LDS and the variance reduction technique of Antithetic Variates used to obtain good quality stable solutions. The problem is solved for each production sequence to determine the sequence with the highest expected profit. The managerial insights are based on extensive numerical experiments on problems of small-size (few products).

The solution approach, which evaluates all possible sequences, is applicable only to problems with a small number of products. The possible number of sequences proliferates with the number of products. In problems with a large number of products, determining the optimal sequence by comparing the expected profit per-unit time of every sequence is computationally expensive. A heuristic approach is proposed for solving a reformulated model which evaluates only a small subset of all possible sequences to provide a near-optimal solution. It is based on the GBD algorithm. The problem reformulation permits the change in production sequence with the change in demand scenarios.

Due to weak lower bounds and slow convergence, it is usual to incorporate techniques to accelerate the conventional GBD algorithm. See Kuthambalayan et al. (2014) and Alkaabneh et al. (2020) for a summary. GBD algorithm's convergence is also improved by combining it with other approaches, for example, the Lagrangian Relaxation algorithm (Hendalianpour et al., 2020).

We tailor the GBD method to our problem and use a non-standard Master and Sub-problem and a heuristic stopping rule based on failure to improve the current best solution over several iterations. We evaluate the performance of the Benders' heuristic against commercial solvers, the DICOPT, and the SBB method to demonstrate its efficacy.

2.3. Studies focusing on pricing and inventory decisions of perishable goods

Although the studies in this section ignore production constraints particular to perishable goods processing, they are still relevant in the sense that they incorporate remaining shelf-life in pricing and inventory decision by:

Modeling demand as a function of price, the current inventory level, and the freshness condition (Agi & Soni, 2020).
Modeling demand as a function of the selling price, reference prices (which the customer finds rational to pay), product freshness, and safety stock (Li & Teng, 2018, Hendalianpour, 2020).
Returning the leftover inventory (Chen et al. 2020) and extending the shelf-life using the renewal process (Shin et al., 2019).
Providing a discount on products nearing the end of their shelf-life (Buisman et al., 2019, Azadi et al., 2019, Krommyda et al., 2020).

Donating the products which exceeded their shelf-life to the food bank to obtain a direct benefit (e.g., tax deduction) and indirect benefits (e.g., goodwill) (Krommyda et al., 2020).
Reducing lost sales using order expedition (Chen et al. 2020) and rescheduling (Georgiadis et al., 2020).
Dynamically adjusting shelf-life as per the product's actual quality either by adjusting the best-before-date or indicating the product quality by other means (Buisman et al., 2019).

2.4. Research positioning

We do not consider overstocking and the risk of dealing with a perishable item after the end of its shelf-life at the retailer. As discussed, the retailer usually manages this risk, when the product is close to the end of its shelf-life, by selling at a discount or donating to the needy or returning to the supplier who may extend shelf-life using a renewal process. The Min-RSL can serve as a signal to the retailer to employ these tactics.

3. Problem description and formulation

In Section 3.1, the problem environment is characterized, and the notations introduced, before providing the mathematical models in Section 3.2 and Section 3.3.

3.1. Notations and assumptions

Order for MTO products is received continuously. Sales accumulate these orders for a time T_1 (Say, from Day 1 morning to Day 3 morning) before communicating it to production. The demand accumulates during one cycle of length T_1 must be satisfied at the end of the subsequent cycle from Day 3 morning till Day 5 morning. This setting is motivated by Lütke Entrup et al. (2005).

The period time T_1, which is also the time of dispatch, determines the capacity available. The available capacity is allocated to the packing of the product. A single product is derived from an intermediated product. Taking the example of yogurt, the intermediate product is the standardized, pasteurized, and homogenized milk, which is fermented. Only filling and packaging (without flavoring) is considered at the pack stage. A single packing material and packet size is assumed, thereby, minor setup is ignored. The major setup is assumed to be sequence-independent. Any natural sequence in the packaging process is neglected. Consider, for example, three products indexed by $i = 1, 2, 3$, which are allotted capacity T_1^L, T_2^L, T_3^L respectively, such that $T_1^L + T_2^L + T_3^L = T_1$.

The sequence of packaging and dispatch time T_1 determine the Max-RSL. If the packaging sequence is $i = 1, 2, 3$, the shelf-life loss of Product 1 at dispatch is $T_1^L + T_2^L + T_3^L$, Product 2 is $T_2^L + T_3^L$, and Product 3 is T_3^L. Let $T_1^C = T_1^L$, $T_2^C = T_1^L + T_2^L$ and $T_3^C = T_1^L + T_2^L + T_3^L$ be the time of packaging completion of the lot, from the start of period T_1, of products 1, 2, and 3. The shelf-life loss of Product 1 is T_1 or T_3^C, Product 2 is $T_1 - T_1^C$, and Product 3 is $T_1 - T_2^C$. A product's shelf-life loss is thus computed from the start of setup (see Fig. 1).

The expected demand per-unit time is given by $D_i(0)$ when the shelf-life loss is zero. The expected demand for Product 1 while considering a reduction in demand with a decrease in remaining shelf-life, is given by

$$\int_{T_1}^{T^{UB}} D_1(0) \times \left(1 - \frac{t}{SL}\right) dt$$

where, SL is shelf-life of the product, and T^{UB} determines the target Min-RSL ($SL - T^{UB}$). Dobson et al. (2017) use a similar demand function.

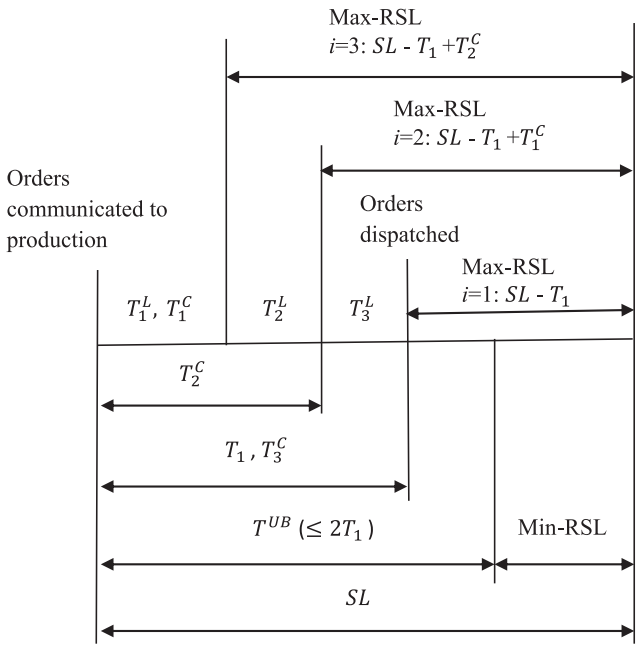


Fig. 1. Maximum and Minimum Remaining Shelf-life in a Production Cycle with production sequence (i = 1,2,3) and each product being setup.

Note that $T_1 \le T^{UB} \le SL$ and $T^{UB} \le 2T_1$. (Though, we consider $T_1 \le T^{UB} \le SL$ we are interested only in situations where $T_1 < T^{UB} < SL$.) The demand for the perishable good is thus not estimated over the sales cycle (from the time of delivery to the time the next replenishment arrives) but until T^{UB}. A positive value of $2T_1 - T^{UB}$ implies less inclination to capture sales, and reduced overstocking risk at the retailer. It provides the retailer with the opportunity to sell all remaining products through marketing promotions, as is usual for perishable goods, before the arrival of the next batch of products or before the end of their shelf-life, whichever is earlier. Retailers usually order less than the sales forecast to achieve a low probability of overstocking (as discussed in Section 1.1).

The expected demand for Product 2 is given by

$$\int_{(T_1 - T_1^C)}^{T^{UB}} D_2(0) \times \left(1 - \frac{t}{SL}\right) dt.$$

The expected demand for Product 3 is given by

$$\int_{(T_1 - T_2^C)}^{T^{UB}} D_3(0) \times \left(1 - \frac{t}{SL}\right) dt.$$

For any product i, the expected demand is given by

$$\int_{(T_1 - T^C_{\{i-1|i>1\}})}^{T^{UB}} D_i(0) \times \left(1 - \frac{t}{SL}\right) dt.$$

Demand uncertainty is captured by adding a stochastic error component to the deterministic demand function. This error component is non-negative with independent uniform distribution and represented by a set of scenarios $E_{il} (l = 1, 2, 3, \cdots, L)$ each with an equal probability of occurrence P_l (Kuthambalayan et al., 2014).

$$D_i = \int_{(T_1 - T^C_{\{i-1|i>1\}})}^{T^{UB}} D_i(0) \times \left(1 - \frac{t}{SL}\right) dt + E_{il}$$

After evaluating the integral, we get

$$D_i = D_i(0) \times \left[T^{UB} \times \left(1 - \frac{T^{UB}}{2 \times SL}\right) - \left(T_1 - T^C_{\{i-1|i>1\}}\right) \times \left(1 - \frac{\left(T_1 - T^C_{\{i-1|i>1\}}\right)}{2 \times SL}\right)\right] + E_{il}$$

The Max-RSL provided to retailers is $SL - T_1 + T^C_{\{i-1|i>1\}}$, and the Min-RSL provided is $SL - T^{UB}$.

In Stage-1, before the demand realization, the operations determines (1) packaging sequence, (2) inventory of intermediate product (I_i^V), (3) capacity T_i^L allocated to packaging product i, (4) time of dispatch (production planning cycle length) T_1, and (5) target Min-RSL based on T^{UB}. In Stage-2, after the demand realization, the packaging quantity Q_{il} of product i in demand scenario l and corresponding unmet demand quantity Q_{il}^S and excess inventory of intermediate product Q_{il}^E, is determined.

The per-unit production cost is C_i and product price is V_i (where $C_i < V_i$). The (sequence-independent) setup time is T_i^S. The unmet demand incurs per-unit penalty C_i^S, and an excess inventory of intermediate product incurs per-unit per-unit-time holding cost H_i. A per-unit per-unit-time product holding cost of C_i^D is incurred on Q_{il} over $(T_1 - T^C_{\{i-1|i>1\}})$. Computing product holding costs in this manner shall induce operations to reduce setup time, lot sizes, and production planning cycle length.

Table 3A
Notation.

Index	
i,j	Index of products $[1, 2, \cdots, N]$
l	Index of demand scenarios $[1, 2, \cdots, L]$
y	Index of sets of L demand scenarios $[1, 2, \cdots, Y]$
Parameter	
L	Number of demand scenarios
N	Number of products
SL	Shelf-life of products (days)
T_i^S	Setup time of product i (days)
R_i	Packaging rate of product i (units/day)
$D_i(0)$	Demand for products i when the shelf-life loss is zero (units/day)
V_i	Per-unit selling price of products i ($/unit)
C_i	Per-unit production cost of products i ($/unit)
C_i^D	Per-unit per-day cost of holding Q_{il} over $(T_1 - T^C_{\{i-1|i>1\}})$ ($/unit/day)
C_i^S	Per-unit penalty on unmet demand of products i ($/unit)
E_{il}	Error component of demand of product i in scenario l. $E_i = U(0, M)$
P_l	Probability of error component of demand in scenario l. $\sum_{l=1}^{L} P_l = 1$
H_i	Per-unit per-day holding cost on the excess units of intermediate product i ($/unit/day)
Decision Variable	
Q_{il}	Quantity of product i packed in scenario l (units)
Q_{il}^S	Unmet demand quantity of product i in scenario l (units)
Q_{il}^E	Excess quantity of intermediate product i in scenario l (units)
T_i^L	Capacity allotted to pack product i (days)
T_j^C	Packaging completion time of product j, measured from the start of production planning cycle
T_1	Common time of dispatch of all products or length of production planning cycle (days)
T^{UB}	Target minimum remaining shelf-life (SL- T^{UB}) (days)
I_i^V	Inventory of intermediate product i at the start of production planning cycle (units)
Binary Variable	
A_i	Setup state of products i { 1 *product i is packed*; 0 *product i is not packed*

The list of indices, parameters, and decisions variable used in the model is summarized in Table 3A.

3.2. Model formulation for pre-defined packaging sequence i = 1,2,..,N (which is not allowed to change with change in demand scenarios)

The objective is to maximize the expected profit per unit time. It comprises five components:

(1) The expected sales-revenue per-unit time:

$$\frac{1}{T_1}\sum_{l=1}^{L} P_l \left(\sum_{i=1}^{N} V_i Q_{il}\right)$$

(2) The expected production-cost per-unit time:

$$\frac{1}{T_1}\sum_{l=1}^{L} P_l \left(\sum_{i=1}^{N} C_i Q_{il}\right)$$

(3) The expected penalty-cost per-unit time of unmet demand:

$$\frac{1}{T_1}\sum_{l=1}^{L} P_l \left(\sum_{i=1}^{N} C_i^S Q_{il}^s\right)$$

(4) The expected holding cost of the product, which is equivalent for each unit in the lot, is imposed from its setup until its dispatch:

$$\frac{1}{T_1}\sum_{l=1}^{L} P_l \left(\sum_{i=1}^{N} C_i^D Q_{il}\left(T_1 - T^C_{\{i-1|i>1\}}\right)\right)$$

(5) The expected holding cost of the excess intermediate product:

$$\sum_{l=1}^{L} P_l \left(\sum_{i=1}^{N} H_i Q_{il}^E\right)$$

The objective function (Z) is the difference in expected sales revenue per-unit time and total expected costs and is provided in Eq. (1).

$$Z = \frac{1}{T_1}\left(\sum_{l=1}^{L} P_l \left(\sum_{i=1}^{N} V_i Q_{il}\right) - \sum_{l=1}^{L} P_l \left(\sum_{i=1}^{N} C_i Q_{il}\right) - \sum_{l=1}^{L} P_l \left(\sum_{i=1}^{N} C_i^S Q_{il}^S\right) - \sum_{l=1}^{L} P_l \left(\sum_{i=1}^{N} C_i^D Q_{il}\left(T_1 - T^C_{\{i-1|i>1\}}\right)\right)\right) - \sum_{l=1}^{L} P_l \left(\sum_{i=1}^{N} H_i Q_{il}^E\right) \quad (1)$$

The objective function is subject to the following constraints:

$$Q_{il} + Q_{il}^S = D_i, \quad i = 1, 2\cdots, N; l = 1, 2\cdots, L \quad (2)$$

$$Q_{il} + Q_{il}^E = I_i^V, \quad i = 1, 2\cdots, N; l = 1, 2\cdots, L \quad (3)$$

$$T_i^S A_i + \frac{Q_{il}}{R_i} \le T_i^L, \quad i = 1, 2\cdots, N; l = 1, 2\cdots, L \quad (4)$$

$$Q_{il} \le M_B \times A_i, \quad i = 1, 2\cdots, N; l = 1, 2\cdots, L \quad (5)$$

$$T_j^C = \sum_{i \le j} T_i^L, \quad i = 1, 2\cdots, N; j = 1, 2\cdots, N \quad (6)$$

$$T_1 = T^C_{\{i|i=N\}} \le T^{UB} \le SL \tag{7}$$

$$T^{UB} \le 2T_1 \tag{8}$$

$$Q_{il}, Q^E_{il}, Q^S_{il}, T^L_i, T^C_i, T^{UB}, I^V_i \ge 0 \forall i, l; A_i \in \{0,1\} \forall i; T_1 \in R_+$$

Constraint (2) ensures that supply (Q_{il}) does not exceed demand (D_i), and the difference between them appears as shortages (Q^S_{il}). Constraint (3) ensures that the packaging quantity (Q_{il}) of the product does not exceed the available intermediate product inventory (I^V_i), and the difference between them appears as excess intermediate product inventory (Q^E_{il}). Constraint (4) ensures that the sum of setup time (T^S_i) and the packaging time $\left(\frac{Q_{il}}{R_i}\right)$ does not exceed the allotted time (T^L_i). Constraint (5) ensures the packaging quantity is zero in case there is not setup, i.e., $A_i = 0$. The value of M_B is fixed at $(D_i(0) + M)$. Constraint (6) determines the packaging completion time T^C_j from the start of the production planning cycle. Constraint (7) determines the time of dispatch T_1, as the packaging completion time $T^C_{\{i|i=N\}}$ of the final product in the production sequence. The time of dispatch does not exceed the T^{UB} which in turn does not exceed the shelf-life SL of the product. Constraint (8) causes the time of the next dispatch of fresh products is no less than T^{UB}.

The operation's objective is to reduce cost by using larger batch sizes and longer production planning cycles (less frequent dispatch). A smaller production planning cycle, with frequent dispatch, improves the freshness of products made available to the customers and requires small lot production. The operation's objective of cost reduction is thus balanced with the marketing objective of providing fresh products.

3.3. Model formulation with packaging sequence allowed to change with change in demand scenarios

Sequencing and capacity allocation can be performed in one single or two separate planning steps. However, since both decisions impact product demand, integrated planning shall prove beneficial. In this section, the model formulated in Section 3.2 is improved with the packaging sequence as one of the decision variables which is permitted to change with the change in demand scenarios. In the Section 4.2, we propose an efficient heuristic based on the Generalized Benders' Decomposition procedure (Benders', 1962; Geoffrion, 1972) for solving the model. Let '*r*' denote the index of the position of a product '*i*' in a production sequence. Assuming that all 'N' products are scheduled once for production, any given product '*i*' can occupy any one of $r = 1,2,..,N$ possible positions in a production sequence. The binary variable A_i, which is used in the model in Section 3.2 (where the production sequence is predefined), is replaced with the binary variable A^M_{ir}. A^M_{ir} is the setup state of products i in position r, given that $\sum_r A^M_{ir} = 1 \forall i$ ('i' occupies only one position in the production sequence) and $\sum_i A^M_{ir} = 1 \forall r$ (a position 'r' in the production sequence is occupied by only one product). The superscript 'M' in A^M_{ir} is meant to improve readability due to the similar notation A_i.

Let B_{ir} denote the capacity allocated to product 'i' in position 'r'. The notation T^L_i, which denotes the capacity allocated to product 'i', is retained in this new formulation. If $A^M_{ir} = 1$, then $B_{ir} = T^L_i$, and if $A^M_{ir} = 0$, then $B_{ir} = 0$ (i.e., $B_{ir} = A^M_{ir} \times T^L_i \forall i, r$). T^C_j is replaced with T^M_r, the packaging completion time in position r, measured from the production planning cycle start (i.e., $T^M_r = T^M_{\{r-1|r>1\}} + \sum_i B_{ir} \forall r$). These additional notations are summarized in Table 3B, followed by the mathematical model.

The objective function (Z) is converted into its minimization form (Z_2) and is given in Equation (1a).

$$Z_2 = -\frac{1}{T_1}\left(\sum_{l=1}^{L} P_l\left(\sum_{i=1}^{N} V_i Q_{il}\right) - \sum_{l=1}^{L} P_l\left(\sum_{i=1}^{N} C_i Q_{il}\right) - \sum_{l=1}^{L} P_l\left(\sum_{i=1}^{N} C^S_i Q^S_{il}\right) - \sum_{l=1}^{L} P_l\left(\sum_{i=1}^{N} C^D_i Q_{il}\left(T_1 - T^C_{\{i-1|i>1\}}\right)\right)\right) + \sum_{l=1}^{L} P_l\left(\sum_{i=1}^{N} H_i Q^E_{il}\right) \tag{1a}$$

The objective function is subject to the following constraints:

$$Q_{il} + Q^S_{il} = D_i, \quad i = 1,2\cdots,N; l = 1,2\cdots,L \tag{2}$$

$$Q_{il} + Q^E_{il} = I^V_i, \quad i = 1,2\cdots,N; l = 1,2\cdots,L \tag{3}$$

$$T^S_i + \frac{Q_{il}}{R_i} \le T^L_i, \quad i = 1,2\cdots,N; l = 1,2\cdots,L \tag{4a}$$

$$T^M_r = T^M_{\{r-1|r>1\}} + \sum_i B_{ir}, \quad r = 1,2\cdots,N \tag{6a}$$

$$T_1 = T^M_{\{r|r=N\}} \le T^{UB} \le SL \tag{7a}$$

$$T^{UB} \le 2T_1 \tag{8}$$

$$\sum_r A^M_{ir} = 1, \quad i = 1,2\cdots,N \tag{9}$$

$$\sum_i A^M_{ir} = 1, \quad r = 1,2\cdots,N \tag{10}$$

$$B_{ir} \le SL \times A^M_{ir}, \quad i = 1,2\cdots,N; r = 1,2\cdots,N \tag{11}$$

$$B_{ir} \le T^L_i, \quad i = 1,2\cdots,N; r = 1,2\cdots,N \tag{12}$$

$$B_{ir} \ge T^L_i + SL \times A^M_{ir} - SL, \quad i = 1,2\cdots,N; r = 1,2\cdots,N \tag{13}$$

$$B_{ir}, Q_{il}, Q^E_{il}, Q^S_{il}, T^L_i, T^M_r, T^{UB}, I^V_i \ge 0 \forall i, r; A^M_{ir} \in \{0,1\} \forall i, r; T_1 \in R_+$$

Assuming each product is setup once, Constraint (4) is re-written as Constraint (4a) without A_i and Constraint (5), which allows the production of a product only when the product is set up on the line, is removed. As discussed later, if it is optimal not to produce a product i, Q_{il} shall take a value close to zero. Constraint (6) is replaced with Constraint (6a), which determines the packaging completion time $\left(T^M_r\right)$ of the product in position 'r' of the packaging sequence. Constraint (7) is re-written as constraint (7a) with $T^M_{\{r|r=N\}}$ replacing $T^C_{\{i|i=N\}}$. Constraint (8) is unchanged.

The Constraint (9) ensures that a product is assigned only one position in the sequence. The Constraint (10) ensures that a position is assigned only one product. Using the relation $T^L_i \le SL \forall i$, the non-linear constraint set $B_{ir} = A^M_{ir} \times T^L_i \forall i, r$ is expressed as a set of three linear Constraints (11)-(13). Constraint (11), (12), and (13) collectively ensure $B_{ir} = T^L_i$, if $A^M_{ir} = 1$ and ensure $B_{ir} = 0$, if $A^M_{ir} = 0$.

Table 3B
Additional Notation.

Index	
r	Index of the position of a product in the production sequence $[1, 2, \cdots, N]$
Decision variable	
B_{ir}	Capacity allocated to product i at position r
T^M_r	Packaging completion time in position r, measured from the production planning cycle start
Binary variable	
A^M_{ir}	$A^M_{ir} = 1$, if product i is assigned to position r, else, $A^M_{ir} = 0$

4. Solution approach

4.1. Sample Average Approximation approach for the model with predefined packaging sequence

In this paper, we assume that the true probability distribution of the error component of the demand can be represented by L discrete scenarios each with a probability of occurrence $P_l = 1/L$. As the number of data points $L \longrightarrow \infty$, the value of the objective function and solution of first-stage decision variables converge to the stochastic programming problem's optimal value (Birge & Louveaux, 2011).

Lemma 1. *For $T_1 \in R_+$, the relaxed 0–1 NLP problem with L discrete scenarios has a unique optimal solution.*

The proof is shown in the Appendix.

The GAMS (General Algebraic Modeling System) is used to model the problem (with L discrete scenarios) and solve using commercial solver SBB.

The stochastic models often use too small scenario tree, one of its reasons is the increase in computational effort with the number of scenarios. The SAA method overcomes this by using random sampling from the true distribution and replication over several iterations to identify candidate solutions. The stochastic error component in the demand function is assumed to have a joint uniform distribution. The demand correlation is considered zero for each pair of products.

There are several methods of generating a limited number of scenarios from a specified continuous distribution. The in-sample stability of the scenario-generation method is confirmed if the optimal values of the objective function in each of the 'Y' replications are approximately the same. The out-of-sample stability means that the value of first-stage decision variables does not depend on which scenario tree we choose (Kaut & Wallace, 2007). (We use the trial and error method initially to arrive at the values of 'L' and 'Y' to ensure that the coefficient of variation, CV = Standard deviation/Mean, of expected profit and the first stage decision variables are approximately 0.01.)

A $L \times N$ matrix of pseudo-random numbers, with $L = 20$, is generated from Gaussian copula with zero correlation parameter to obtain random samples, which are *U(0, 1)*. The error scenarios are obtained by multiplying the $L \times N$ matrix by the joint uniform distribution's maximum value (say, 100). The problem is solved 'Y = 30 (y = 1,2,..,Y)' times using a different set of *'L'* scenarios each time. The upper-bound on expected profit is then given by $\overline{Z} = (1/Y)\sum_y Z_y$, and the standard deviation is computed as $\sqrt{\sum_y (\overline{Z} - Z_y)^2/Y}$.

Also, a N dimensional Sobol's LDS is sequentially generated $L \times Y$ times and multiplied by the joint uniform distribution's maximum value. (LDS, also known as 'quasi-random' sequences, are numbers; in comparison to pseudo-random numbers, they are better equidistributed in a given volume). The problem is solved 'Y' times using a different set of *'L'* scenarios each time. The upper-bound on expected profit is then given by $\overline{Z}_1 = (1/Y)\sum_y Z_{1y}$, and the standard deviation is computed as $\sqrt{\sum_y (\overline{Z}_1 - Z_{1y})^2/Y}$. We use Sobol's LDS based on a comparison of stability with the pseudo-random numbers. Please refer to Table 4A, in which the solution reported is for parameters value at High Level and Set A of R_1, R_2, R_3 (refer to Table 5) for sequence $i = 1, 3, 2$.

In addition, the variance-reduction technique of Antithetic Variates is adopted to further improve the solution stability. The original values of Sobol's LDS are subtracted from 'one' to obtain '$L \times Y$' additional N-dimensional values, which is multiplied by the joint uniform distribution's maximum value. The problem is solved additional 'Y' times using these new set of *'L'* scenarios each time to obtain $\overline{Z}_2 = \left(\frac{1}{Y}\right)\sum_y Z_{2y}$. The upper-bound on expected profit is then given by $\overline{Z} = \left(\frac{1}{2}\right)\sum_y (\overline{Z}_1 + \overline{Z}_2)$ and the standard deviation computed as:

$$\sqrt{\left(\sum_y (\overline{Z}_1 - Z_{1y})^2 + \sum_y (\overline{Z}_2 - Z_{2y})^2 + 2\sum_y (\overline{Z}_1 - Z_{1y})(\overline{Z}_2 - Z_{2y})\right)/4Y}$$

4.2. Generalized Benders' Decomposition approach for the model, which allows packaging sequence to change with change in demand scenarios

The number of possible sequences for a 3 product problem is only 3! or six. By comparing the expected profit per-unit time of every sequence (with a single set of demand scenarios or by using the SAA approach), the optimal sequence with the highest expected profit per-unit time can be obtained. The solution approach, which evaluates all possible sequences, is applicable only to problems with a small number of products. The possible number of sequences proliferates with the number of products. For example, a 7 product problem has 7! or 5040, a 9 product problem has 9! or 362880, and a 11 product problem has 11! or 39,916,800 possible sequences. In problems with many products, determining the optimal sequence by comparing the expected profit per-unit time of every sequence is computationally prohibitive.

A heuristic approach to solve practical size problems is proposed in this section, which evaluates only a small subset of all possible sequences, corresponding to a single set of demand scenarios, to provide a near-optimal solution. It is based on the Generalized Benders' Decomposition (GBD) algorithm. The GBD algorithm solves complex MINLP problems by decomposing them into two smaller problems: a MIP Master problem and an NLP Sub-problem. The method first solves the MIP problem then solves the NLP problem. The dual value of relevant constraint(s) of the NLP problem is used to derive an optimality cut at each iteration, which is added to the MIP problem. Both the Master problem and Sub-problem are iteratively solved. In a minimization problem, the MIP solution provides the lower-bound to the problem and

Table 4A
Stability of scenario-generation method ($A_i = \{0, 1, 1\}$ in each solution).

	Pseudo-random		Sobol's LDS		Antithetic Variates	
	Mean	CV	Mean	CV	Mean	CV
Z	65112.4	0.027	64059.7	0.010	64012.9	0.008
T_1	2.2026	0.029	2.2410	0.012	2.2437	0.008
T^{UB}	3.2602	0.060	3.2990	0.022	3.2994	0.016
I_2^V	69.3278	0.077	71.3098	0.036	71.5521	0.012
I_3^V	192.06	0.045	194.168	0.014	194.055	0.009
T_2^L	1.0146	0.071	1.0416	0.033	1.0449	0.011
T_3^L	1.1880	0.039	1.1995	0.013	1.1989	0.008

Table 4B
Notation for Benders' Decomposition Heuristic.

Index	
k	Number of iterations $[1, 2, \cdots, K]$
Parameter	
f^*_{3irk}	Optimal value of dual variable corresponding to Constraint (11a) in iteration k
f^*_{4irk}	Optimal value of dual variable corresponding to Constraint (13a) in iteration k
A^{M*}_{irk}	Optimal value of A^M_{ir} in iteration k
A^{M*}_{ir}	Value of A^M_{ir} used to construct sub-problem
A^{M**}_{ir}	Optimal value of A^M_{ir} at termination of heuristic
I^{V**}_i	Optimal value of I^V_i at termination of heuristic
Z^*_{2k}	Objective function value of sub-problem in iteration k

is non-decreasing with each iteration. The best solution to the NLP problem, at any iteration, provided the upper-bound. Motivated by the GBD method, the two-stage MINLP problem is partitioned into a non-standard MIP Master problem and NLP Sub-problem-1. The additional notation used in the Benders' Heuristic is provided in Table 4B.

The NLP Sub-problem-1 comprises the Objective function (1a). The objective function is subject to the constraint sets (2), (3), (4a), (6a), (7a), (8). The Constraints (11), (12) and (13) are replaced with the Constraints (11a), (12a), and (13a) respectively.

$$B_{ir} \le SL \times A_{ir}^{M^*}, \quad i = 1,2\cdots,N; r = 1,2\cdots,N \tag{11a}$$

$$B_{ir} \le T_i^L, \quad i = 1,2\cdots,N; r = 1,2\cdots,N \tag{12a}$$

$$B_{ir} \ge T_i^L + SL \times A_{ir}^{M^*} - SL, \quad i = 1,2\cdots,N; r = 1,2\cdots,N \tag{13a}$$

The dual value of Constraint (11a) and (13a) are used to add Optimality cut to the master problem.

The Master problem:

$$\min Z_1 = \theta \tag{14}$$

$$\theta \ge \sum_{i|A_{irk-1}^{M^*}=0 r|A_{irk-1}^{M^*}=0} \sum f_{3irk-1}^{*} A_{ir}^{M} + \sum_{i|A_{irk-1}^{M^*}=0 r|A_{irk-1}^{M^*}=0} \sum f_{4irk-1}^{*} A_{ir}^{M}, \quad k = 2,3\cdots K \tag{15}$$

$$\sum_r A_{ir}^M = 1, \quad i = 1,2\cdots,N \tag{9}$$

$$\sum_i A_{ir}^M = 1, \quad r = 1,2\cdots,N \tag{10}$$

$$\sum_{i|A_{irk-1}^{M^*}=0 r|A_{irk-1}^{M^*}=0} \sum A_{ir}^M \ge 1, \quad k = 2,3,..K \tag{16}$$

$$A_{ir}^M \in \{0,1\} \forall i,r; i = 1,2\cdots,N; r = 1,2\cdots,N$$

The Constraint (15) is the optimality cut added to the master problem. It provides lower bound to the value of θ. Constraint (16) ensures that the packaging sequence at each MIP solution differs from those used to construct previous sub-problems.

The Benders' Heuristic repetitively solves the Master problem and the Sub-problem-1. The objective function of the Sub-problem-1 with L discrete scenarios is quasi-convex, and the constraint set convex (the proof is similar to that shown in Appendix). In the first iteration, the value of the complicating variable (A_{ir}^M) is fixed such that the Constraints (9) and (10) are not violated. The Sub-problem-1 is then solved. The dual value of Constraint (11a) and (13a) of the NLP Sub-problem-1 is used to construct the optimality cut. Second iteration onwards, the Master problem's solution, with the optimality cut, gives the value of complicating variables, which is used to solve the Sub-problem-1. The Master problem uses the optimality cut to learn each complicating variable influence on the problem solution. This eventually leads to an identification of the optimal complicating vector. The MIP Master problem is solved using the CPLEX solver, and the NLP Sub-problem-1 is solved using the CONOPT solver.

The Master problem is not the standard GBD version and does not provide the lower-bound on the objective function value. The Sub-problem-1 allows production setups with zero production and does not provide the true upper-bound on the objective function value. The proposed heuristic is terminated if there is no improvement in the upper-bound over twenty consecutive iterations. The value of A_{ir}^M at termination (i.e., $A_{ir}^{M^{**}}$) and the corresponding value of I_i^V (i.e., $I_i^{V^{**}}$) are recorded.

A Sub-problem-2 is then solved with Objective function (1) subject to Constraints (2), (3), (7a) and (8), and Constraints (4a), (6a), (11), (12) and (13) is replaced with the Constraints (4b), (6b), (17) and (18):

$$T_i^L \le 0 \ for \ \left\{i = 1,2\cdots,N | I_i^{V^{**}} \le w\right\} \tag{17}$$

$$Q_{il} \le 0 \ for \ \left\{i = 1,2\cdots,N; l = 1,2\cdots,L | I_i^{V^{**}} \le w\right\} \tag{18}$$

$$T_i^S + \frac{Q_{il}}{R_i} \le T_i^L \ for \ \left\{i = 1,2\cdots,N; l = 1,2\cdots,L | I_i^{V^{**}} > w\right\} \tag{4b}$$

$$T_r^M = T_{\{r-1|r>1\}}^M + \sum_i A_{ir}^{M^{**}} \times T_i^L, r = 1,2\cdots,N \tag{6b}$$

Constraint (17) and (18) ensures that if the value of I_i^V is negligible (in our case, $w = 0.00001$) at the termination of the algorithm, then $T_i^L = 0$ and $Q_{il} = 0$ in the solution of Sub-problem-2. Thus the assumption that each product is setup once is relaxed. Constraint (4b) ensures that if the value of I_i^V is not negligible, T_i^L allocated is no less than the sum of setup time and packaging time of the lot. Constraint (6b) computes packaging completion time of the product in position 'r' of the sequence (considered from the start of the production planning cycle).

The objective function of the Sub-problem-2 with L discrete scenarios is quasi-concave, and the constraint set convex (the proof is similar to that shown in Appendix). The Benders' Heuristic is summarized in the Appendix.

5. Computational result and discussion

5.1. Computational results of SAA approach

This section performs sensitivity analysis on key problem parameters, namely, demand variance, product holding cost, production rate, intermediate product holding cost, setup time, and shortage cost, to analyze the effect on expected profit per-unit time. The computational study captures the interaction between key decision variables under the integrated approach, which leads to better decisions.

5.1.1. Experiment design

We consider an example of two ($N = 2$), three products ($N = 3$), and four products ($N = 4$) with possible number of sequences as 2!, 3!, and 4!, respectively. For example, the possible packaging sequences with $N = 3$ are (1,3,2), (3,1,2), (3,2,1), (1,2,3), (2,1,3) and (2,3,1). The value of shortage cost (C_i^S), product holding cost (C_i^D), setup time (T_i^S), production rate (R_i), and intermediate product holding cost (H_i) vary across product type i. Two sets of values: Low and High, are considered for these parameters. This setting is motivated by Holmbom & Segerstedt (2014) and values are set using trial and error method to ensure that $T_1 < T^{UB} < SL$. The values of V_i, C_i, and D_i (0) are fixed. The Yogurt's shelf-life is about 8–10 days (Lütke Entrup et al., 2005), and we considered the shelf-life of the product to be ten days in this study.

The demand variance E_{il} is considered uniformly distributed *U(0, 40)*, and the higher level is considered *U(0, 100)*. The experiment design is depicted in Table 5 and results in 128 (2^7) experiments and is repeated for each of the production sequence. Two, three, and four products result in 2, 6, and 24 possible sequences. The problem is solved for all possible sequences, leading to 4096 (128 × (2 + 6 + 24)) experiments. Moreover, the result of each experiment is based on 60 replications of demand scenarios. So, the problem is solved overall at 2, 45, 760 (4096 × 60) times.

5.1.2. Sensitivity analysis

In this section, the results with $N = 3$ are provided. (The results with $N = 2$ and $N = 4$ are provided in a supplementary file and support the findings summarized in this section.) The problem is solved for all six possible sequences. In Table 6A, for Set A of R_1, R_2, R_3, the profit per-unit time of sequence $(1,3,2)$ is highest in all settings. The setup state is $\{0,1,1\}$, implying the Product 1 is not packed. In Table 6B, for Set B of R_1, R_2, R_3, the profit per-unit time of sequence (1,2,3) is the highest (the setup state is $\{0,1,1\}$), except at setting $C_i^S = HL, R_i = LL, T_i^S = LL, E_{il} = HL, C_i^D = HL$, and $H_i = HL$, when (2,1,3) is highest. Tables 6A and 6B

Table 5
Experiment design.

	Values					
N	2		3		4	
L	20		20		20	
V_i	800, 3200		800, 2560, 3200		800, 2560, 3200, 2186	
C_i	500, 2000		500, 1600, 2000		500, 1600, 2000, 1366	
$D_i(0)$	63, 102		31, 90, 51		31, 90, 51, 57	
SL	10		10		10	
Factor	***LL***	***HL***	***LL***	***HL***	**Low level (*LL*)**	**High level (*HL*)**
C_i^S	100, 400	200, 800	100, 320, 400	200, 640, 800	100, 320, 400, 273	200, 640, 800, 546
C_i^D	25, 100	50, 200	25, 80, 100	50, 160, 200	25, 80, 100, 68	50, 160, 200, 136
H_i	20, 80	40, 160	20, 64, 80	40, 128, 160	20, 64, 80, 55	40, 128, 160, 110
E_{il}	$U(0, 40)$	$U(0, 100)$	$U(0, 40)$	$U(0, 100)$	$U(0, 40)$	$U(0, 100)$
T_i^S	0.0357, 0.0357	0.0714, 0.0714	0.0357, 0.0714, 0.0357	0.0714, 0.1428, 0.0714	0.0357, 0.0714, 0.0357, 0.0476	0.0714, 0.1428, 0.0714, 0.0952
R_i **-SetA**	84, 42	118, 59	84, 105, 42	147, 183, 73	147, 184, 74, 134	210, 262, 105, 192
R_i **-SetB**	105, 84	147, 118	105, 42, 84	183, 73, 147	184, 74, 147, 134	262, 105, 210, 192

Table 6A
Expected profit per-unit time, and (T_1, T^{UB}) at High level of H_1, H_2, H_3 for the optimal production sequence $i = 1, 3, 2$ for Set A and $N = 3$.

(E_{il})			*LL*		*HL*	
(C_i^D)			*LL*	*HL*	*LL*	*HL*
(C_i^S)	(R_i)	(T_i^S)				
LL	*LL*	*LL*	59782.3 (1.98, 2.65)	52476.6 (1.43, 1.81)	51679.0 (2.76, 3.68)	41545.2 (2.0, 2.38)
LL	*LL*	*HL*	56545.0 (2.34, 3.17)	48261.9 (1.69, 2.15)	49365.7 (3.04, 4.10)	38622.2 (2.20, 2.62)
LL	*HL*	*LL*	105393.8 (1.75, 3.07)	94818.5 (1.26, 2.06)	96125.4 (2.30, 4.17)	82134.2 (1.65, 2.58)
LL	*HL*	*HL*	99152.0 (2.17, 3.94)	86593.8 (1.57, 2.54)	91303.2 (2.64, 4.94)	75867.6 (1.90, 2.98)
HL	*LL*	*LL*	54700.9 (2.36, 3.11)	46387.2 (1.71, 2.06)	43435.0 (3.46, 4.56)	31780.5 (2.51, 2.71)
HL	*LL*	*HL*	52031.5 (2.67, 3.59)	42953.1 (1.94, 2.32)	41669.4 (3.69, 4.92)	29600.1 (2.69, 2.94)
HL	*HL*	*LL*	98583.9 (2.03, 3.53)	86674.1 (1.47, 2.29)	85111.4 (2.82, 4.99)	68895.2 (2.04, 2.98)
HL	*HL*	*HL*	93239.7 (2.40, 4.30)	79703.5 (1.73, 2.72)	81294.5 (3.12, 5.78)	64012.9 (2.24, 3.29)

Table 6B
Expected profit per-unit time, and (T_1, T^{UB}) at High level of H_1, H_2, H_3 for the optimal production sequence $i = 1, 2, 3$ for Set B and $N = 3$.

(E_{il})			*LL*		*HL*	
(C_i^D)			*LL*	*HL*	*LL*	*HL*
(C_i^S)	(R_i)	(T_i^S)				
LL	*LL*	*LL*	51945.2 (1.83, 2.25)	46544.7 (1.35, 1.59)	45113.3 (2.57, 3.12)	37293.2 (1.91, 2.13)
LL	*LL*	*HL*	49377.9 (2.18, 2.69)	43307.5 (1.61, 1.89)	43302.7 (2.83, 3.47)	35090.3 (2.10, 2.35)
LL	*HL*	*LL*	88052.1 (1.61, 2.45)	80586.5 (1.18, 1.68)	80899.7 (2.09, 3.17)	70854.6 (1.53, 2.06)
LL	*HL*	*HL*	83108.1 (2.04, 3.17)	74212.7 (1.49, 2.12)	77021.6 (2.43, 3.75)	65972.1 (1.78, 2.44)
HL	*LL*	*LL*	47364.5 (2.24, 2.73)	41008.9 (1.65, 1.90)	37476.0 (3.37, 3.98)	*28339.7** (2.68, 2.71)
HL	*LL*	*HL*	45312.8 (2.53, 3.12)	38466.6 (1.87, 2.14)	36129.8 (3.54, 4.28)	26604.5 (2.62, 2.77)
HL	*HL*	*LL*	82189.1 (1.91, 2.88)	73539.9 (1.40, 1.94)	71073.5 (2.65, 3.99)	58912.5 (1.95, 2.54)
HL	*HL*	*HL*	78054.1 (2.27, 3.50)	68255.3 (1.66, 2.31)	68118.9 (2.92, 4.45)	55277.0 (2.15, 2.81)

*: Optimal production sequence is $i = 2, 1, 3$ and setup state is {1, 1, 1}.

provides the change in expected profit per-unit time for a high level of H_1, H_2, H_3, for the optimal production sequence, as parameter values are varied. The production planning cycle length (T_1) and the value of T^{UB} (which determines the Min-RSL) are provided within the parenthesis.

An increase in demand variability and shortage cost reduce the expected profit per-unit time due to increased aggregate cost. Setup time introduces idleness in the production line and decreases the supply capability. Thus an increase in setup time decreases the expected profit per-unit time. To counteract this, the operations increase the time of dispatch T_1 to increase the supply capability. In a joint decision case, the marketing counters the reduction in Max-RSL and product demand by increasing T^{UB}, but at the expense of Min-RSL. Moreover, an increase in T_1 also reduces the Max-RSL at the time of dispatch, thus capturing the trade-off between operational costs and customer service in terms of product freshness. Integrated operation-marketing models thus capture the complex trade-offs through feedback control, which sequential decision-making would have overlooked.

The expected product holding cost is computed over $T_1 - T^C_{\{i-1|i>1\}}$, where T_1 is the time of dispatch and $T^C_{\{i-1|i>1\}}$ is the time at which its setup is started. An increase in product holding cost increases the aggregate cost and consequently decreases the expected profit per unit time. To counteract this, operations reduces the time of dispatch T_1. In joint decision making, the marketing reduces T^{UB} to decrease the product demand, which helps the operations to reduce the time of dispatch T_1. Moreover, with a decrease in T_1 and T^{UB}, the Max-RSL and Min-RSL are increased, and customer service in terms of product freshness is improved.

An increase in production rate increases the supply capability as well as the expected profit per-unit time. To utilize the supply potential, operations decreases T_1. The Max-RSL at the time of dispatch thus increases. With low product holding cost and high setup time, the downward pressure on T_1 is low, or cycle length reduces to a lesser extent, and in a joint decision case, marketing increases T^{UB} and pursues more sales by reducing Min-RSL.

5.1.3. Benefits of integrating sequencing decisions with capacity allocation decision

In Table 7A, for Set A of R_1, R_2, R_3, the profit per-unit time of sequence (1, 3, 2) is the highest. The remaining sequences arranged in descending order of the profit per-unit time are: (3,1,2), (3,2,1), (1,2,3),

Table 7A
Maximum and Minimum benefits of integrating sequencing decisions in percentage at High level of H_1, H_2, H_3 for Set A and $N = 3$.

(E_{il})			LL		HL	
(C_i^D)			LL	HL	LL	HL
(C_i^S)	(R_i)	(T_i^S)				
LL	LL	LL	(28.4, 1.7)	(32.5, 2.0)	(30.9, 1.9)	(39.0, 2.5)
LL	LL	HL	(28.9, 1.7)	(33.8, 2.1)	(31.4, 1.9)	(40.3, 2.7)
LL	HL	LL	(19.5, 1.1)	(22.1, 1.3)	(20.8, 1.2)	(24.9, 1.4)
LL	HL	HL	(20.0, 1.1)	(23.2, 1.3)	(21.2, 1.2)	(26.0, 1.5)
HL	LL	LL	(35.5, 3.8)	(42.8, 4.8)	(41.7, 4.6)	(57.8, 6.3)
HL	LL	HL	(36.1, 3.9)	(44.4, 5.0)	(42.3, 4.6)	(60.1, 7.0)
HL	HL	LL	(23.6, 2.5)	(27.6, 2.9)	(26.5, 2.7)	(34.3, 3.7)
HL	HL	HL	(24.1, 2.5)	(28.9, 3.1)	(26.9, 2.8)	(35.5, 3.8)

Table 7B
Maximum and Minimum benefits of integrating sequencing decisions in percentage at High level of H_1, H_2, H_3 for Set B and $N = 3$.

(E_{il})			LL		HL		
(C_i^D)			LL	HL	LL	HL	
(C_i^S)	(R_i)	(T_i^S)					
LL	LL	LL	(38.3, 2.6)	(43.1, 2.8)	(40.8, 2.7)	(48.6, 3.0)	
LL	LL	HL	(39.5, 2.7)	(45.6, 3.1)	(42.2, 2.8)	(51.4, 3.5)	
LL	HL	LL	(27.2, 1.8)	(30.5, 2.0)	(28.7, 1.8)	(33.6, 2.0)	
LL	HL	HL	(28.3, 1.9)	(33.1, 2.1)	(29.8, 1.9)	(35.9, 2.2)	
HL	LL	LL	(39.1, 2.0)	(44.9, 1.8)	(43.9, 0.8)	(58.9, 0.23)	*
HL	LL	HL	(40.5, 2.8)	(47.6, 2.8)	(45.0, 1.4)	(61.7, 0.21)	**
HL	HL	LL	(28.8, 2.1)	(33.1, 2.2)	(30.9, 1.2)	(37.8, 0.6)	
HL	HL	HL	(30.4, 2.9)	(36.0, 3.3)	(32.4, 1.9)	(40.8, 1.7)	

* The best, second-best and worst sequence are $i = 2, 1, 3$; $i = 1, 2, 3$ and $i = 3, 1, 2$ repectively.
**The best, second-best and worst sequence are $i = 1, 2, 3$; $= 2, 1, 3$; and $i = 3, 1, 2$ repectively.

(2,1,3) and (2,3,1). The difference between the expected profit per-unit time with the optimal packaging sequence (1, 3, 2) and that with the worst packaging sequence (2, 3, 1) (the second-best sequence (3, 1, 2)) provides the maximum (the minimum) benefit of integrating sequencing decisions. These values are expressed as maximum ($Benefit_1$) and minimum ($Benefit_2$) percentage benefits. Table 7A provides the change in these benefit percentages for high H_1, H_2, H_3 levels as parameter values are varied. We aim to identify conditions in which it is best to integrate sequencing and capacity allocation decisions.

In Table 7B, for Set B of R_1, R_2, R_3, the profit per-unit time of sequence $(1, 2, 3)$ is the highest. The remaining sequences arranged in descending order of the profit per-unit time are: (2,1,3), (2,3,1), (1,3,2), (3,1,2) and (3,2,1). At setting $C_i^S = HL, R_i = LL, T_i^S = HL, E_{il} = HL, C_i^D = HL$ and $H_i = HL$, however, sequences in descending order of profit per-unit time are: $(1, 2, 3)$, $(2, 1, 3)$, $(2, 3, 1)$, $(1, 3, 2)$, $(3, 2, 1)$ and $(3, 1, 2)$. At setting $C_i^S = HL, R_i = LL, T_i^S = LL, E_{il} = HL, C_i^D = HL$ and $H_i = HL$, sequences in descending order of profit per-unit time are: (2,1,3), (1,2,3), (2,3,1), (1,3,2), (3,2,1) and (3,1,2). These values of maximum ($Benefit_1$) and minimum ($Benefit_2$) percentage benefit are computed as before. Table 7B provides the change in these benefit percentages for high H_1, H_2, H_3 levels as parameter values are varied.

An increase in product holding cost, demand variance, and shortage costs increase the maximum benefit of integrating sequencing decisions with capacity allocation. $Benefit_1^{HL} = \frac{Z_{Best}^{HL} - Z_{Worst}^{HL}}{Z_{Best}^{HL}}$, $Benefit_1^{LL} = \frac{Z_{Best}^{LL} - Z_{Worst}^{LL}}{Z_{Best}^{LL}}$ and $Benefit_1^{HL} > Benefit_1^{LL}$ implies that $(1 - \frac{Z_{Worst}^{HL}}{Z_{Best}^{HL}}) > (1 - \frac{Z_{Worst}^{LL}}{Z_{Best}^{LL}})$ or $(1 - \frac{Z_{Worst}^{HL}}{Z_{Worst}^{LL}}) > (1 - \frac{Z_{Best}^{HL}}{Z_{Best}^{LL}})$. The percentage reduction in expected profit with an increase of product holding cost, demand variance, and shortage costs is less in the integrated approach. So these conditions promote simultaneous sequencing and capacity allocation.

An increase in production rate reduces the maximum benefit of integration. $Benefit_1^{HL} < Benefit_1^{LL}$ implies that $(1 - \frac{Z_{Worst}^{HL}}{Z_{Best}^{HL}}) < (1 - \frac{Z_{Worst}^{LL}}{Z_{Best}^{LL}})$ or $(1 - \frac{Z_{Best}^{LL}}{Z_{Best}^{HL}}) < (1 - \frac{Z_{Worst}^{LL}}{Z_{Worst}^{HL}})$. The percentage increase of expected profit with an increase in production rate is less in the case of integrating sequencing and capacity allocation decisions. A lower production rate thus promotes the integration of sequencing decision. The small values of minimum benefit indicate a minor penalty in expected profit per-unit time on the selection of second best, instead of the best, product sequence (for example, due to production constraints). The summary of results is provided in Table 8.

5.2. Computational result of Benders' heuristic

We solve the problem, which allows the packaging sequence to change with change in demand scenarios, with 2, 3, and 4 products, using Benders' Heuristic and SBB solver, with a single set of demand scenarios. The results are presented in Tables 16A–16B, 17A–17B, and 18A–18B as supplementary data. For 2, 3, and 4 products, the problem instances solved with the parameter settings specified in Table 5 are 128 each. The expected profit per-unit time obtained using Benders' heuristic compares well with the optimal value provided by the SBB solver. For the 2, 3, and 4 product problems the average (maximum) percent optimality gap is 0.87% (6.72%), 0.69% (5.48%), and 0.53% (6.70%), respectively. The optimal sequence obtained with a single set of demand scenarios using the SBB solver differs in several instances, from that of the problem, which applies the SAA approach to evaluate each possible packing sequence. In the case of 2, 3, and 4 product problems, it differs in 56, 69, and 76 instances.

We also consider a large-size problem of seven $(N = 7)$, nine $(N = 9)$, and eleven $(N = 11)$ products. The error component in demand for each product is considered as $U(0, 100)$, and $L \times N$ matrices of pseudo-random numbers are generated. The probability value $(1/L)$ for each error scenario is the same. Two such sets $(Y = 2)$ of random numbers are generated to evaluate the change in production sequence with the change in demand scenarios. Ten instances of the seven products problem and five instances of the nine and eleven product problem are solved using the proposed heuristic. Ten sets of parameters (one for each instance) are generated using the uniform distribution for the 7 product problems and five sets of parameters for the 9 and 11 product problems, whose details are provided in Table 9.

In the case of seven, nine, and eleven products, there are 7! (i.e., 5040), 9! (i.e., 362880), and 11! (i.e., 39916800) possible sequences, respectively. One of these sequences is used as a starting solution to solve the NLP Sub-problem-1, by setting $A_{ir}^{M^*} = 1$ for $i = r$ and $A_{ir}^{M^*} = 0$ for $i \neq r$. For benchmarking the proposed heuristic's performance, the ten instances of the problem with seven products, each solved twice with a different set of demand scenarios, are also solved using the DICOPT

Table 8
Summary of Results.

	Profit (Z)	(T_1)	(T^{UB})	$Benefit_1$
$C_i^D(\uparrow)$	↓	↓	↓	↑
$R_i(\uparrow)$	↑	↓	↑*	↓
$E_{il}(\uparrow)$	↓	↑	↑	↑
$C_i^s(\uparrow)$	↓	↑	↑	↑
$T_i^s(\uparrow)$	↓	↑	↑	
$H(\uparrow)$	↓		↑	

*: at LL of C_i^D and HL of T_i^S.

Table 9
Test environment.

Parameter	Value	Parameter	Value
		(Twenty sets)	
N	7, 9, 11	C_i ($/unit)	$U(500, 2000)$
L	50	C_i^S ($/unit)	$U(200, 800)$
SL	10	C_i^D ($/unit/day)	$U(50, 200)$
(Two set: Y = 2)		D_i (units/day)	$U(15, 45)$
E_i (units/day)	$U(0, 100)$	H_i ($/unit/day)	$U(40, 160)$
		R_i (units/day)	$U(115, 288)N = 7U(126, 315)N = 9$
			$U(168, 420)N = 11$
		T_i^S (days)	$U(0.0357, 0.0714)$
		V_i ($/unit)	$U(800, 3200)$

solver and SBB solver.

The problem is modeled using GAMS 24.7.3 on a desktop computer with Pentium Core 2 Duo processor, 2 gigabytes RAM and running Windows 7 Professional x32. The Master problem is solved using the CPLEX solver (with a relative gap of zero), and the Sub-problem is solved using the CONOPT solver. SBB solution is terminated with a relative gap of 5%, and the DICOPT solution is terminated as soon as NLP sub-problem solution starts deteriorating.

The results of the ten instances of the seven product problems, each solved twice with a different set of demand scenarios, are provided in Table 10. Expressly, the objective value, number of NLP problems solved, and CPU seconds are provided. The percentage gap in the DICOPT solvers objective value and Benders' Heuristic method with respect to the objective value of the SBB solver is computed for each problem instance.

The mean, best and worst gap in objective function value relative to that of SBB for the:

1. DICOPT are 29.2 %, 5.88%, and 63.3%, respectively, and
2. Benders' Heuristic are 1.80%, 0, and 5.76%, respectively.

The SBB solution indicates that the production sequence changes with the change in demand scenario. However, the minor difference indicates a small penalty in expected profit per-unit time in the choice of an alternate near-optimal production sequence.

The mean, maximum and minimum numbers of NLP solved for the:

SBB are 50415, 98170, and 20140, respectively,
DICOPT are 5, 6, and 3, respectively, and
Benders' Heuristic are 52, 79, and 29, respectively.

The mean, maximum and minimum CPU seconds for the:

SBB method are 17535.4, 35134.1, and 5152.5,respectively,
DICOPT method are 444.6, 810.5, and 209.4, respectively, and
Benders' Heuristic are 13.89, 24.74, and 6.20, respectively.

The computational difficulty of solving the SMINLP problem with only seven products is reflected in the CPU seconds required by the SBB solver to terminate with a 5% relative gap. In nine products, the SBB method required 76759.6 (199914) mean, 167999.4 (328320) maximum, and 49865.7 (166440) minimum CPU seconds (NLP solves)

Table 10
Comparison with DICOPT and SBB solvers for $N = 7$.

	y = 1					y = 2				
Method	Z	NLP No.	Gap %	Sequence	CPU second	Z	NLP No.	Gap %	Sequence	CPU second
SBB	50477.8	32,708	–	(7,5,6,4,3,1,2)	9314.8	50125.4	59,594	–	(7,5,6,4,3,1,2)	33516.2
DICOPT	43930.7	5	12.9	(7,5,1,3,6,4,2)	675.4	43832.9	5	12.5	(7,5,1,3,2,4,6)	729.9
Benders'	48775.9	71	3.37	(4,7,6,5,3,1,2)	22.15	48440.2	71	3.36	(4,7,6,5,3,1,2)	23.01
SBB	135625.2	32,669	–	(3,6,7,5,4,2,1)	7941.5	135090.8	67,987	–	(3,6,7,5,4,1,2)	12474.8
DICOPT	124974.1	5	7.85	(5,6,7,4,2,3,1)	344.6	124270.1	4	8.00	(5,6,7,2,3,1,4)	222.6
Benders'	131166.9	40	3.28	(6,4,5,7,2,1,3)	11.61	130467.1	40	3.42	(6,4,5,7,2,1,3)	11.21
SBB	96347.0	92,383	–	(6,5,3,7,1,2,4)	22647.7	95985.3	98,170	–	(6,5,3,7,4,1,2)	31255.5
DICOPT	37020.1	3	61.5	(5,3,2,4,1,7,6)	219.2	36471.9	3	62.0	(5,3,1,6,7,2,4)	209.4
Benders'	94451.1	79	1.96	(7,5,3,6,2,1,4)	23.79	94109.6	79	1.95	(7,5,3,6,2,1,4)	24.74
SBB	139199.2	41,606	–	(4,7,6,3,1,2,5)	11247.0	137888.6	38,948	–	(4,7,6,5,1,3,2)	10880.3
DICOPT	131004.2	5	5.88	(6,7,4,1,5,2,3)	440.7	129581.0	4	6.00	(6,7,4,3,1,5,2)	346.7
Benders'	139199.2	29	0	(4,7,6,2,1,3,5)	6.20	137888.6	29	0	(4,7,6,2,1,3,5)	6.68
SBB	86819.1	84,084	–	(5,3,7,6,2,1,4)	23771.4	85578.1	64,346	–	(5,3,7,6,2,1,4)	35134.1
DICOPT	74923.3	4	13.7	(7,3,5,2,6,1,4)	539.5	71894.0	4	15.9	(7,4,3,6,2,5,1)	428.2
Benders'	84294.2	40	2.90	(6,4,5,7,2,1,3)	9.96	83270.6	40	2.69	(6,4,5,7,2,1,3)	9.15
SBB	106183.8	21,416	–	(7,5,4,1,2,6,3)	13101.9	105522.4	51,299	–	(7,5,4,6,3,2,1)	24156.3
DICOPT	39109.3	4	63.1	(4,2,7,5,6,1,3)	437.4	38711.1	6	63.3	(4,6,7,2,5,1,3)	810.5
Benders'	106183.8	31	0	(7,5,4,1,3,2,6)	8.00	105522.4	31	0	(7,5,4,1,3,2,6)	8.45
SBB	129479.0	23,028	–	(6,7,3,5,2,1,4)	10782.0	129053.3	29,668	–	(6,7,3,5,2,1,4)	11876.5
DICOPT	49546.6	3	61.7	(7,5,3,1,4,2,6)	255.4	47992.7	3	62.8	(7,5,3,4,1,2,6)	215.7
Benders'	122399.2	41	5.46	(6,4,5,7,2,1,3)	9.45	121609.7	41	5.76	(6,4,5,7,2,1,3)	9.66
SBB	152212.2	61,606	–	(7,5,4,6,1,2,3)	11811.5	151504.2	20,140	–	(7,5,4,6,2,1,3)	5152.5
DICOPT	121529.3	4	20.1	(5,6,4,7,1,2,3)	297.1	120522.3	4	20.4	(5,6,2,7,1,3,4)	351.1
Benders'	151694.1	39	0.34	(6,5,4,7,2,1,3)	6.28	150998.6	39	0.33	(6,5,4,7,2,1,3)	6.50
SBB	79627.4	39,812	–	(3,5,6,7,4,2,1)	15484.6	79528.0	65,788	–	(3,5,6,7,4,2,1)	23546.2
DICOPT	69283.0	4	12.9	(7,5,2,3,6,4,1)	379.9	69164.5	4	13.0	(7,5,2,3,1,4,6)	440.6
Benders'	79083.2	79	0.68	(7,5,3,6,2,1,4)	21.17	78984.8	79	0.6	(7,5,3,6,2,1,4)	21.57
SBB	112038.1	30,773	–	(6,7,4,5,3,1,2)	16778.9	111482.7	52,260	–	(6,7,4,5,3,2,1)	19835.9
DICOPT	77872.0	5	30.4	(7,5,4,6,2,1,3)	747.4	77266.8	6	30.6	(7,5,2,6,4,3,1)	801.7
Benders'	112024.0	71	0.01	(4,7,6,5,3,1,2)	21.18	111482.0	71	0	(4,7,6,5,3,1,2)	21.11

Table 11
Comparison with SBB solver (solution restricted to 10,000 CPU seconds) for $N = 9$ and $N = 11$.

	y = 1					y = 2				
	Z	NLP No.	RG %	Sequence	cpu sec	Z	NLP No.	RG %	Sequence	cpu sec
$N = 9$										
SBB	90,409	32,433	85	(2,8,7,5,9,6,3,4,1)	–	84,192	66,975	80	(8,5,3,4,6,9,7,2,1)	–
BH	87,900	33	–	(7,8,3,6,1,5,9,4,2)	10.1	**86,647**	33	–	(7,8,3,6,1,5,9,4,2)	10.5
SBB	91,934	30,038	85	(8,5,7,6,4,2,1,9,3)	–	91,063	44,884	80	(7,8,4,5,6,2,1,3,9)	–
BH	90,797	34	–	(8,7,3,6,1,5,9,4,2)	9.8	90,997	34	–	(8,7,3,6,1,5,9,4,2)	10.0
SBB	88,239	28,310	83	(6,4,8,5,9,2,3,7,1)	–	84,747	31,189	84	(8,1,6,4,9,5,7,3,2)	–
BH	86,824	32	–	(8,4,3,6,1,5,9,7,2)	9.3	**85,151**	32	–	(8,4,3,6,1,5,9,7,2)	9.3
SBB	104,316	34,697	82	(1,7,4,3,9,8,2,6,5)	–	104,574	53,176	77	(4,7,2,1,9,8,3,6,5)	–
BH	100,955	33	–	(7,8,3,6,1,5,9,4,2)	9.1	98,621	33	–	(7,8,3,6,1,5,9,4,2)	9.6
SBB	88,648	38,714	86	(8,5,7,2,4,9,1,6,3)	–	86,962	42,185	82	(2,5,8,9,7,1,6,4,3)	–
BH	85,512	43	–	(4,5,2,3,8,7,9,1,6)	13.7	83,849	43	–	(4,5,2,3,8,7,9,1,6)	13.0
$N = 11$										
SBB	110,352	25,570	81	(4,11,10,7,1,8,2,9,3,5,6)	–	113,221	26,225	80	(6,4,9,11,10,7,8,1,2,5,3)	–
BH	105,217	30	–	(8,4,2,10,1,5,11,7,3,9,6)	16.6	104,069	30	–	(8,4,2,10,1,5,11,7,3,9,6)	18.0
SBB	107,812	25,857	81	(6,4,9,10,7,5,11,3,8,2,1)	–	116,768	27,751	80	(5,7,9,8,10,6,4,11,1,2,3)	–
BH	106,296	55	–	(8,4,9,1,6,5,3,7,10,2,11)	31.3	107,797	55	–	(8,4,9,1,6,5,3,7,10,2,11)	30.8
SBB	93,181	26,651	86	(5,8,6,9,2,7,10,11,3,1,4)	–	86,865	25,621	86	(9,6,8,10,3,2,7,4,1,5,11)	–
BH	**98,102**	46	–	(5,6,9,2,8,10,3,7,1,4,11)	27.5	**99,142**	46	–	(5,6,9,2,8,10,3,7,1,4,11)	25.0
SBB	101,070	24,991	83	(11,6,5,4,8,9,3,1,2,7,10)	–	92,714	26,488	84	(6,5,8,4,11,9,7,1,2,10,3)	–
BH	**105,092**	36	–	(4,5,6,1,11,7,9,2,10,3,8)	20.2	**103,379**	36	–	(4,5,6,1,11,7,9,2,10,3,8)	19.6
SBB	106,467	28,282	82	(9,11,10,6,4,8,5,2,7,3,1)	–	103,879	27,577	82	(4,8,10,1,9,6,11,5,3,7,2)	–
BH	**107,202**	27	–	(8,4,6,10,1,5,11,2,3,9,7)	14.8	**105,123**	27	–	(8,4,6,10,1,5,11,2,3,9,7)	14.8

to solve five instances of the problem with a 5% relative gap, for y = 1. (The mean, best and worst gaps in objective function value relative to SBB for the Benders' Heuristic are 5.14%, 3.53%, and 6.72%, respectively.) Moreover, the CPLEX solver could not solve the MIP subproblem in the DICOPT method with a 1 million node limit, and the relative gap at termination was still 97%. Due to the excessive computation effort, the DICOPT solution is not provided. For the nine and eleven product problems, a limit of 10,000 CPU seconds is used in the SBB method and five (instead of ten) instances are solved for each of the two sets of demand scenarios. The results of the five instances, each solved twice with a different set of demand scenarios, are provided in Table 11. In the table, $RG = (Z_{BB} - Z_{BI})/Z_{BB}$, where Z_{BB} is the maximum value of the objective for the sub-nodes that have not been solved yet and Z_{BI} is the value of the objective for the best integer solution found so far. Based on its solution quality and CPU seconds at termination, the Benders' heuristic (BH) capability to solve large size problems can be judged.

6. Conclusion and directions for future research

Studies on the planning of Make-and-Pack production of perishable products, such as yogurt, tend to focus on the packaging stage. The packaging stage is a bottleneck resource and is capital intensive. The traditional mathematical models for production planning are modified to account for perishability risks in the formulation explicitly, in our case, a MTO strategy (with order penetration point at the inventory of intermediated product), remaining shelf-life dependent demand, and an age-dependent expected product holding cost component in the objective function. In addition, the target Min-RSL that marketing/retailer wishes to provide to the consumers is included in the demand function.

We study an aggregate level planning problem and integrate sequencing decisions. The purpose is to evaluate the integrated operations-marketing firm's ability to balance two conflicting objectives: reducing costs and improving customer service in terms of product freshness. Additionally, we aim to gain insights into conditions that promote the integration of sequencing and capacity allocation decisions.

Towards this end, we propose a two-stage stochastic MINLP model, for a pre-determined packaging sequence. The first-stage aggregate level operations decisions on the: (1) starting inventory of each intermediate product, (2) length of production planning cycle (limited by the products shelf-life), which determines the Max-RSL, and (3) capacity allocation to each product, and the marketing decision on the Min-RSL. In the second stage, market demand is realized. The product demand decreases from the Max-RSL to the Min-RSL, and the period is limited by the length of the production planning cycle. A stochastic error component captures the uncertainty in demand. The intermediate product inventory is allocated to implement the packaging plan and meet the demand of MTO products. The model developed captures trade-offs related to the inventory of intermediate products, shortages, and remaining shelf-life dependent customer demand.

The SMINLP problem is converted into its deterministic equivalent using the SAA method. The stochastic error component of demand is thereby represented using a set of scenarios with an equal probability of occurrence. The demand scenarios are represented using Sobol's LDS, and the stability of the solutions improved using the variance reduction technique of Antithetic Variates. The problem is solved for each of the possible production sequences to identify the optimal sequence, second-best, and the worst sequence in terms of expected profit per-unit time.

Sensitivity analysis on key problem parameters, namely, demand variance, product holding cost, production rate, intermediate product holding cost, setup time, and shortage cost, provides insights into the behavior of the expected profit per-unit time for the optimal sequence. The proposed integrated operation-marketing model can balance the conflicting objectives of reducing costs and improving customer service in terms of product freshness through feedback control and provide ideal solutions. We restrict our attention to cases where Min-RSL is strictly less than the Max-RSL. In an integrated-firm, the operations challenges of high setup time or demand variability or demand satisfaction requirement (shortage costs) result in a lower Max-RSL and reduced demand. Marketing counters by reducing the Min-RSL and trying to capture sales. A higher value of product holding cost signifies the importance of customer service in product freshness and results in increases in both Max-RSL and Min-RSL. An increase in packaging rate causes Max-RSL to gains, and marketing, under low product holding cost and high setup times, lowers Min-RSL. Moreover, the integrated sequencing and capacity allocation setting, represented by the optimal production sequence, and sequential decision-making, represented by the worst sequence, are compared. Numerical comparisons show that high demand variability, shortage cost, and product holding cost, and low packaging rate promote integrated decision-making. There is a minor

penalty in expected profit per-unit time on the selection of second best, instead of the best, product sequence.

The model is improved to permit the change in production sequence with the change in demand scenarios. The solution approach, which evaluates all possible sequences, is applicable only to problems with a small number of products. An efficient heuristic based on the GBD method is proposed to solve problems in practice. The heuristic evaluates only a small subset of all possible sequences to provide a near-optimal solution. The solution of the proposed Benders' Heuristic method and the DICOPT solver is compared to that of the SBB solver on several large problem instances. The proposed method performs better than the DICOPT solver in terms of solution quality. In terms of the number of NLP solves, the proposed method is significantly improved over the SBB method and compares well with the DICOPT method. In terms of computational effort, it performs better than both these commercial solvers.

This study highlights the importance of coordination at the Marketing-Operations interface and the need to understand the complex trade-offs seen in practice. Future research could incorporate: (1) the management of perishability risk, when the product is close to the end of its shelf-life (refer to studies in Section 2.2) or when Min-RSL is exceeded, (2) inventory routing decisions, see Wang et al. (2018) and Alkaabneh et al. (2020), and (3) supply disruption risks, see Yavari et al. (2020) who use dynamic pricing and routing decision and Diabat et al. (2019) who design supply network, considering this aspect.

CRediT authorship contribution statement

Nitesh Kumar Singh: Software, Validation, Formal analysis, Investigation, Data curation, Visualization. **Thyagaraj S. Kuthambalayan:** Writing – original draft, Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Proof Lemma 1. In the MINLP problem, the Objective function (1) is a sum of non-linear functions and a linear function. The domain of the decision variables in the non-linear part (f) of the objective function is restricted to $Q_{il} \geq 0$, $Q_{il}^{S} \geq 0$, $T_i^D \geq 0$, and $T_1 > 0$ (where $T_i^D = T_1 - T^C_{\{i-1|i>1\}}$).

$$f = \frac{1}{T_1}\left(Q_{il}V_i - Q_{il}C_i - Q_{il}^{s}C_i^{S} - T_i^{D}C_i^{D}Q_{il}\right)$$

We have,

$$f_1 = -\frac{1}{(T_1)^2}\left(Q_{il}V_i - Q_{il}C_i - Q_{il}^{s}C_i^{S} - Q_{il}T_i^{D}C_i^{D}\right), f_2 = -\frac{1}{T_1}\left(Q_{il}C_i^{D}\right), f_3 = -\frac{1}{T_1}\left(C_i^{S}\right)$$

$$f_4 = \frac{1}{T_1}\left(V_i - C_i - T_i^{D}C_i^{D}\right), f_{11} = \frac{2}{(T_1)^3}\left(Q_{il}V_i - Q_{il}C_i - Q_{il}^{s}C_i^{S} - Q_{il}T_i^{D}C_i^{D}\right), f_{12} = \frac{1}{(T_1)^2}\left(Q_{il}C_i^{D}\right)$$

$$f_{13} = \frac{1}{(T_1)^2}\left(C_i^{S}\right), f_{14} = -\frac{1}{(T_1)^2}\left(V_i - C_i - T_i^{D}C_i^{D}\right), f_{21} = \frac{1}{(T_1)^2}\left(Q_{il}C_i^{D}\right), f_{22} = 0, f_{23} = 0$$

$$f_{24} = -\frac{1}{T_1}\left(C_i^{D}\right), f_{31} = \frac{1}{(T_1)^2}\left(C_i^{S}\right), f_{32} = 0, f_{33} = 0, f_{34} = 0, f_{41} = -\frac{1}{(T_1)^2}\left(V_i - C_i - T_i^{D}C_i^{D}\right)$$

$f_{42} = -\frac{1}{T_1}\left(C_i^{D}\right), f_{43} = 0$ and $f_{44} = 0$.

Where,

$f_1 = \frac{\partial f}{\partial T_1} f_2 = \frac{\partial f}{\partial T_i^{D}} f_3 = \frac{\partial f}{\partial Q_{il}^{s}} f_4 = \frac{\partial f}{\partial Q_{il}} f_{11} = \frac{\partial^2 f}{\partial (T_1)^2} f_{12} = \frac{\partial^2 f}{\partial T_1 \partial T_i^{D}} f_{13} = \frac{\partial^2 f}{\partial T_1 \partial Q_{il}^{s}} f_{14} = \frac{\partial^2 f}{\partial T_1 \partial Q_{il}} f_{21} = \frac{\partial^2 f}{\partial T_i^{D} \partial T_1} f_{22} = \frac{\partial^2 f}{\partial (T_i^{D})^2} f_{23} = \frac{\partial^2 f}{\partial T_i^{D} \partial Q_{il}^{s}} f_{24} = \frac{\partial^2 f}{\partial T_i^{D} \partial Q_{il}} f_{31} = \frac{\partial^2 f}{\partial Q_{il}^{s} \partial T_1} f_{32} = \frac{\partial^2 f}{\partial Q_{il}^{s} \partial T_i^{D}}$,

$f_{33} = \frac{\partial^2 f}{\partial (Q_{il}^{s})^2} f_{34} = \frac{\partial^2 f}{\partial Q_{il}^{s} \partial Q_{il}} f_{41} = \frac{\partial^2 f}{\partial Q_{il} \partial T_1} f_{42} = \frac{\partial^2 f}{\partial Q_{il} \partial T_i^{D}} f_{43} = \frac{\partial^2 f}{\partial Q_{il} \partial Q_{il}^{s}}$ and $f_{44} = \frac{\partial^2 f}{\partial (Q_{il})^2}$.

Since,

$$B_1 = \begin{vmatrix} 0 & f_1 \\ f_1 & f_{11} \end{vmatrix} \leqslant 0, B_2 = \begin{vmatrix} 0 & f_1 & f_2 \\ f_1 & f_{11} & f_{12} \\ f_2 & f_{21} & f_{22} \end{vmatrix} \geqslant 0, B_3 = \begin{vmatrix} 0 & f_1 & f_2 & f_3 \\ f_1 & f_{11} & f_{12} & f_{13} \\ f_2 & f_{21} & f_{22} & f_{23} \\ f_3 & f_{31} & f_{32} & f_{33} \end{vmatrix} \leqslant 0, B_4 = \begin{vmatrix} 0 & f_1 & f_2 & f_3 & f_4 \\ f_1 & f_{11} & f_{12} & f_{13} & f_{14} \\ f_2 & f_{21} & f_{22} & f_{23} & f_{24} \\ f_3 & f_{31} & f_{32} & f_{33} & f_{34} \\ f_4 & f_{41} & f_{42} & f_{43} & f_{44} \end{vmatrix} \geqslant 0.$$

For a non-linear function to be quasi-concave in non-negative orthant it is necessary that the leading principal minor $|B_1| \leqslant 0$, $|B_2| \geqslant 0$, $|B_3| \leqslant 0 \cdots\cdots\cdots |B_n| \leq 0 (\geq 0)$ when n is odd (even) (Chiang & Wainwright, 2005). Thus, it is proved that function f is quasi-concave and so is the objective function.

Moreover, except Constraint (2) all constraints are linear.

$$Q_{il}+Q_{il}^{S}=D_i(0)\times\left[T^{UB}\times\left(1-\frac{T^{UB}}{2SL}\right)-\left(T_1-T^{C}_{\{i-1|i>1\}}\right)\times\left(1-\frac{\left(T_1-T^{C}_{\{i-1|i>1\}}\right)}{2SL}\right)\right]+E_{il}.$$

After substituting $T_1-T^{C}_{\{i-1|i>1\}}=T_{11i}$ and simplification, the Constraint (2) can be written as,

$$Q_{il}+Q_{il}^{S}=D_i(0)\times T^{UB}-D_i(0)\times\frac{\left(T^{UB}\right)^2}{2SL}-D_i(0)\times T_{11i}+D_i(0)\times\frac{(T_{11i})^2}{2SL}+E_{il}.$$

After further simplification it can be written as,

$$Q_{il}+Q_{il}^{S}+\frac{D_i(0)}{2SL}\times\left(\left(T^{UB}\right)^2-(T_{11i})^2\right)=D_i(0)\times T^{UB}-D_i(0)\times T_{11i}+E_{il}$$

$$Q_{il}+Q_{il}^{S}+\frac{D_i(0)}{2SL}\times\left(T^{UB}-T_{11i}\right)\times\left(T^{UB}+T_{11i}\right)=D_i(0)\times\left(T^{UB}-T_{11i}\right)+E_{il}$$

After substituting $T^{UB}-T_{11i}=T_{12i}$ and $T^{UB}+T_{11i}=T_{13i}$, the Constraint (2) is expressed as

$$Q_{il}+Q_{il}^{S}+\frac{D_i(0)}{2SL}\times T_{12i}\times T_{13i}=D_i(0)\times T_{12i}+E_{il}.$$

Let the non-linear term be denoted by $W_i=T_{12i}\times T_{13i}$. Then by taking natural log on both sides, we get

$$\ln W_i=\ln T_{12i}+\ln T_{13i}$$

Let $F=\ln T_{12i}+\ln T_{13i}-\ln W_i=0$, which implicitly defines T_{12i} as a function of $T_{13i} For T_1\in R_+ and T_1\le T^{UB}, T_{13i}>0$. By implicit function rule and log rule, for positive value of T_{12i},

$$\frac{dT_{12i}}{dT_{13i}}=-\frac{\partial F/dT_{13i}}{\partial F/dT_{12i}}=-\frac{1/T_{13i}}{1/T_{12i}}=-\frac{T_{12i}}{T_{13i}}<0.$$

Then it follows that,

$$\frac{d^2T_{12i}}{dT_{13i}^{\,2}}=\frac{d}{dT_{13i}}\left(\frac{dT_{12i}}{dT_{13i}}\right)=\frac{d}{dT_{13i}}\left(-\frac{T_{12i}}{T_{13i}}\right)=-\frac{1}{(T_{13i})^2}\left[T_{13i}\left(\frac{dT_{12i}}{dT_{13i}}\right)-T_{12i}\right]\rangle 0.$$

The sign of the first derivative and second derivative establishes any isoquant to be downward sloping throughout and strictly convex for a positive value of T_{12i} and T_{13i}. Thus, the constraint set is convex.

Benders' Heuristic

Step 0 (Initialization)

1. $k=1, UB_{k-1}=\infty$
2. Fix $A_{ir}^{M^*}$ to satisfy constraints (9) and (10).

Step 1 (Solve the NLP Sub-problem-1)

1. Solve the NLP Sub-problem-1
2. Return f^*_{3irk} and f^*_{4irk}
3. Update $UB_k=\min\left(Z^*_{2k}, UB_{k-1}\right)$
4. If $(UB_k<UB_{k-1})$
   Store: A_{ir}^{M}, I_i^{V}
5. If $((k>20\ and\ UB_k=UB_{k-1}\ and\ UB_k=UB_{k-2}\cdots\cdots\ and\ UB_k=UB_{k-20})$ or (if the Master problem is infeasible)): STOP.
   Return stored values: $A_{ir}^{M^{**}}=A_{ir}^{M}$ and $I_i^{V^{**}}=I_i^{V}$
   Solve the NLP Sub-problem-2
   Return $Z_2, T^{UB}, T_1, T_i^{L}, T_r^{M}, I_i^{V}$ as near-optimal values

Step 2 (Add optimality cut)

1. Add optimality cut (15) to the Master problem using $f^*_{3irk}.f^*_{4irk}$

Step 3 (Solve the Master problem)

1. k = k + 1
2. Solve the Master problem
3. Return to Step 1 with $A_{ir}^{M^*}$

Appendix B. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cie.2021.107812.

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